

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value
1	Response Time (Avg)	< 2 seconds	1, 18 seconds
2	Response Time (Max)	5 seconds	2.86 seconds
3	Throughput (Req/sec)		1487 req/sec
4	Error Rate		0.32%
5	CPU Utilization		63%
6	Memory Utilization		56%

Step 4: Observations & Analysis

Key Findings:

The system responded within the acceptable time range for both average and maximum response time.

Throughput was stable and the system handled concurrent user load efficiently. Error rate remained below 1% indicating high reliability.

Bottlenecks Identified:

- High database query time during peak concurrent requests.
- Some delay observed in loading dashboard analytics due to heavy data processing.

Optimization Steps Taken:

- Optimized SQL queries and added indexing on frequently accessed tables,
- Implemented caching for dashboard data to reduce response time.
- Enabled connection pooling to improve database performance,

Step 5: Screenshots | Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

[Insert Screenshot 1 here — JMeter Summary Report]

[Insert Screenshot 2 here — Response Time Graph]